



MorphoScan Pro - Setup Guide

Complete setup guide for developers to get the MorphoScan Pro project up and running locally.



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Prerequisites

Before you begin, ensure you have the following installed:

Required Software

- **Node.js:** v20.x or v22.x (tested with v22.14.0)
- **npm:** v10.x or higher (tested with v10.9.2)
- **Git:** Latest version

Optional Software

- **Docker:** For local Supabase instance (optional)
- **Supabase CLI:** For database management (optional)
- **Stripe CLI:** For testing webhooks locally (optional)

Verify Installation

```
node --version    # Should show v20.x or v22.x
npm --version     # Should show v10.x or higher
git --version     # Should show latest version
```



Quick Start

Get up and running in 5 minutes:

```
# 1. Clone the repository (if not already cloned)
git clone <repository-url>
cd visionary_scanner_suite

# 2. Install dependencies
npm install

# 3. Set up environment variables
cp .env.example .env.local
# Edit .env.local with your configuration

# 4. Start development server
npm run dev

# 5. Open in browser
# Visit http://localhost:8080
```

Detailed Setup

Step 1: Clone Repository

```
git clone <repository-url>
cd visionary_scanner_suite
```

Step 2: Install Dependencies

```
npm install
```

This will install:

- **1,130+ packages** including React 18, TypeScript 5, Vite 7
- Frontend frameworks: Tailwind CSS, Radix UI, Framer Motion
- Backend integration: Supabase, Stripe
- Development tools: ESLint, Prettier, Playwright, Vitest

Expected output:

```
added 1130 packages, and audited 1131 packages in 3m
found 0 vulnerabilities 
```

Step 3: Environment Configuration

Using .env.local (Recommended for Development)

```
# Copy the local development template
cp .env.local .env.local.backup
# Or copy from example
cp .env.example .env.local
```

Edit .env.local

Open `.env.local` and configure:

```
# Supabase Configuration (provided or use your own)
VITE_SUPABASE_URL=https://thajylrvfzjmerqqkmjv.supabase.co
VITE_SUPABASE_PUBLISHABLE_KEY=<your-key-here>
VITE_SUPABASE_PROJECT_ID=thajylrvfzjmerqqkmjv

# App Configuration
VITE_APP_ENV=development
VITE_APP_VERSION=sfw
VITE_DISTRIBUTION_CHANNEL=direct

# Contact Information
VITE_SUPPORT_CONTACT_EMAIL=support@yourapp.com
VITE_PRIVACY_CONTACT_EMAIL=privacy@yourapp.com
ADMIN_SUPER_EMAIL=admin@yourapp.com

# Security
VITE_CLIENT_ENCRYPTION_SALT=your-unique-salt-here
```

Important Notes:

- ✓ `.env.local` takes precedence over `.env`
- ✓ Never commit `.env.local` to version control
- ✓ Use different salts for dev/staging/production
- ✓ Stripe keys are optional for basic development

Database Setup

Option 1: Use Existing Supabase Instance (Recommended)

The project is configured to use a Lovable Cloud Supabase instance. No additional setup needed!

Option 2: Local Supabase with Docker

If you want a fully local development environment:

Prerequisites

- Docker Desktop installed and running

Setup Steps

```
# 1. Install Supabase CLI (if not already installed)
npm install -g supabase

# 2. Start local Supabase instance
npm run db:start

# This will:
# - Start PostgreSQL database
# - Start Supabase Studio (UI)
# - Start Auth, Storage, and Edge Functions
# - Apply all 98 migrations automatically
```

Local Supabase URLs

Once started, update your `.env.local` :

```
VITE_SUPABASE_URL=http://localhost:54321  
VITE_SUPABASE_PUBLISHABLE_KEY=<get-from-cli-output>
```

Database Management Commands

```
# Start local database  
npm run db:start  
  
# Stop local database  
npm run db:stop  
  
# Reset database (WARNING: Deletes all data)  
npm run db:reset  
  
# Push schema changes  
npm run db:push  
  
# Generate TypeScript types  
npm run db:types
```

Database Migrations

The project includes **98 comprehensive migrations** covering:

- ☒ User authentication & profiles
- ☒ Subscription management (Stripe integration)
- ☒ DLC (Downloadable Content) system
- ☒ Health tracking & analytics
- ☒ Sexual wellness features
- ☒ Community & social features
- ☒ Achievement & referral systems
- ☒ Privacy settings & data retention
- ☒ Admin management tools
- ☒ Push notifications & device tokens

Migration Files Location: `supabase/migrations/`

First Migration: `20241209000001_dlc_system.sql`

Latest Migration: `20251226023338_289e405f-58d6-4fdb-a222-d7b50ab2f708.sql`



Running the Application

Development Mode

```
npm run dev
```

Server Information:

- Local: `http://localhost:8080`
- Network: `http://[your-ip]:8080`

Features enabled in dev mode:

- ☒ Hot Module Replacement (HMR)
- ☒ Fast refresh for React components
- ☒ Source maps for debugging
- ☒ TypeScript type checking
- ☒ ESLint warnings

Production Preview

```
# Build for production
npm run build

# Preview production build
npm run preview
```

Specialized Build Commands

```
# Build different app versions
npm run build:sfw          # Safe-for-work version
npm run build:nsfw         # Adult content version
npm run build:hybrid       # Both modes available

# Build for different distribution channels
npm run build:sfw:store    # SFW for App Store
npm run build:sfw:direct   # SFW for direct download
npm run build:nsfw:direct  # NSFW for direct download

# Analyze bundle size
npm run build:analyze
```

 Build & Deployment**Build for Production**

```
npm run build
```

Build Output:

- Output directory: `dist/`
- Build time: ~40-45 seconds
- Bundle size: ~6 MB (compressed)
- PWA support: Enabled with service worker

Build includes:

- ☒ Optimized and minified assets
- ☒ Code splitting and lazy loading
- ☒ Service worker for offline support
- ☒ Progressive Web App (PWA) manifest
- ☒ Static asset optimization

Build Warnings

You may see warnings about large chunks:

```
(!) Some chunks are larger than 1500 kB after minification
```

This is expected for:

- 3D model viewer (916 KB)
- jsPDF library (385 KB)
- Chart libraries (392 KB)

These will be optimized in Phase 3 (Performance Optimization).



Testing

Unit & Integration Tests

```
# Run tests
npm test

# Run tests in UI mode
npm test:ui

# Run tests once (CI mode)
npm run test:run

# Generate coverage report
npm run test:coverage
```

End-to-End Tests

```
# Run E2E tests
npm run test:e2e

# Run E2E tests in UI mode
npm run test:e2e:ui

# Debug E2E tests
npm run test:e2e:debug
```

Run All Tests

```
npm run test:all
```

Code Quality

Linting

```
# Check for linting errors
npm run lint

# Fix linting errors automatically
npm run lint:fix
```

Formatting

```
# Check code formatting
npm run format

# Format code automatically
npm run format:write
```

Pre-commit Hooks

The project uses Husky for pre-commit hooks:

-  Automatic linting on staged files
-  Automatic formatting on commit
-  TypeScript type checking

Troubleshooting

Common Issues & Solutions

Issue: Dependencies won't install

Symptoms:

```
npm ERR! code ERESOLVE
npm ERR! ERESOLVE unable to resolve dependency tree
```

Solutions:

```
# Clear npm cache
npm cache clean --force

# Delete node_modules and package-lock.json
rm -rf node_modules package-lock.json

# Reinstall
npm install

# If still failing, try legacy peer deps
npm install --legacy-peer-deps
```

Issue: Build fails with TypeScript errors

Symptoms:

```
error TS2307: Cannot find module
```

Solutions:

```
# Regenerate TypeScript types from Supabase
npm run db:types

# Check TypeScript configuration
npx tsc --noEmit

# Clear Vite cache
rm -rf node_modules/.vite
npm run build
```

Issue: Supabase connection errors**Symptoms:**

```
Error: Invalid Supabase URL
AuthApiError: Invalid authentication credentials
```

Solutions:

1. Verify `.env.local` has correct values
2. Check Supabase project status at <https://app.supabase.com>
3. Ensure API keys are not expired
4. Verify network connectivity

```
# Test Supabase connection
curl https://thajylrvfzjmerqqkmjv.supabase.co/rest/v1/
```

Issue: Port already in use**Symptoms:**

```
Error: Port 8080 is already in use
```

Solutions:

```
# Find process using port 8080
lsof -i :8080

# Kill the process
kill -9 <PID>

# Or use a different port
PORT=3000 npm run dev
```

Issue: Out of memory during build**Symptoms:**

```
JavaScript heap out of memory
```


Solutions:

```
# Increase Node.js memory limit
NODE_OPTIONS="--max-old-space-size=4096" npm run build

# Or add to package.json scripts:
"build": "NODE_OPTIONS='--max-old-space-size=4096' vite build"
```

Issue: Hot Module Replacement (HMR) not working**Symptoms:**

- Changes not reflecting in browser
- Page requires manual refresh

Solutions:

```
# Clear Vite cache
rm -rf node_modules/.vite

# Restart dev server
npm run dev

# Check file watcher limits (Linux)
echo fs.inotify.max_user_watches=524288 | sudo tee -a /etc/sysctl.conf
sudo sysctl -p
```



Development Workflow

Recommended Development Flow

1. Start Development Server

```
bash
npm run dev
```

2. Make Changes

- Edit files in `src/`
- Changes auto-reload via HMR

3. Test Your Changes

```
bash
npm test
npm run lint
```

4. Build Before Committing

```
bash
npm run build
```

5. Commit Changes

```
bash
git add .
git commit -m "feat: your changes"
# Pre-commit hooks run automatically
```

Project Structure

```
visionary_scanner_suite/
├── src/
│   ├── components/      # React components
│   ├── hooks/           # Custom React hooks
│   ├── lib/             # Utilities & helpers
│   ├── pages/           # Page components
│   ├── integrations/    # Supabase & external services
│   └── config/          # App configuration
├── public/              # Static assets
├── supabase/
│   ├── migrations/      # Database migrations
│   └── functions/       # Edge functions
├── docs/                # Documentation
├── scripts/             # Build & deployment scripts
└── e2e/                 # End-to-end tests
```



Important Notes

Security

- ⚠️ **Never commit `.env` or `.env.local` files**
- ⚠️ Use different encryption salts per environment
- ⚠️ Keep Stripe secret keys server-side only
- ⚠️ Rotate API keys regularly

Performance

- Large chunks (>1500 KB) are expected for:
- 3D model viewer
- PDF generation
- Chart libraries
- Code splitting optimization planned in Phase 3

Database

- 98 migrations must be run in order
- Migrations are automatically applied when using Supabase CLI
- For production, ensure migrations are run before deployment



Additional Resources

Documentation

- [QUICK_START_GUIDE.md](#) (./QUICK_START_GUIDE.md) - User-facing quick start
- [DEPLOYMENT_CHECKLIST.md](#) (./DEPLOYMENT_CHECKLIST.md) - Pre-deployment checklist
- [PRODUCTION_TESTING_GUIDE.md](#) (./PRODUCTION_TESTING_GUIDE.md) - Production testing
- [README.md](#) (./README.md) - Project overview

Package Scripts Reference

```
# Development
npm run dev           # Start dev server
npm run build         # Build for production
npm run preview       # Preview production build

# Testing
npm test             # Run unit tests
npm run test:e2e     # Run E2E tests
npm run test:all     # Run all tests

# Code Quality
npm run lint         # Check linting
npm run lint:fix     # Fix linting errors
npm run format       # Check formatting
npm run format:write # Format code

# Database
npm run db:start     # Start local Supabase
npm run db:stop      # Stop local Supabase
npm run db:reset     # Reset database
npm run db:types     # Generate TS types

# Specialized Builds
npm run build:sfw    # Build SFW version
npm run build:nsfw   # Build NSFW version
npm run build:analyze # Analyze bundle
```

Setup Checklist

Use this checklist to ensure everything is configured correctly:

- ☐ Node.js v20+ or v22+ installed
- ☐ npm v10+ installed
- ☐ Repository cloned
- ☐ Dependencies installed (`npm install`)
- ☐ `.env.local` created and configured
- ☐ Supabase connection verified
- ☐ Development server starts (`npm run dev`)
- ☐ Production build works (`npm run build`)
- ☐ Tests pass (`npm test`)
- ☐ Linting passes (`npm run lint`)
- ☐ Database migrations understood
- ☐ Git hooks working (Husky)

Ready to Develop!

You're all set! Start the development server:

```
npm run dev
```

Visit <http://localhost:8080> and start building! 🚀

Support

- **Issues:** Open a GitHub issue
 - **Email:** support@morphoscan.dev
 - **Documentation:** See `/docs` folder
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Phase: 1/6 - Setup & Environment Configuration